

Research Proposal

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Preliminary title:

Coercion and Vulnerability: District-Level Land Use Consolidation under Rwanda's Crop Intensification Program and Smallholder Resilience to Shocks.

Research question:

How does district-level intensity of land use consolidation under Rwanda's Crop Intensification Program shape household dietary resilience to adverse shocks, and does this effect vary by household wealth?

Sub-questions:

- Is district-level LUC intensity negatively associated with household dietary diversity scores among households that experienced an adverse shock in the past 12 months?
- Does this vulnerability-amplifying effect increase as household wealth decreases, indicating that the poorest households bear the greatest resilience cost?

Current knowledge:

Puzzle and relevance. Rwanda's Crop Intensification Program (CIP), launched in 2007, mandated smallholder farmers to cultivate state-assigned monocrops on consolidated plots. Compliance was enforced through Rwanda's *imihigo* performance contract system—a cascade of accountability running from the president down to individual farmers (Cioffo et al., 2016). The programme produced nationally reported yield increases and is widely cited as a demonstration of what strong state capacity can achieve in agricultural development (Cantore, 2011).

Yet this aggregate productivity narrative conceals a distributional question: *who paid the price for these gains?* Del Prete et al. (2019) find that land consolidation increased production of roots and tubers while reducing dietary diversity. Clay and Zimmerer (2020) go further, arguing that the CIP eroded smallholder *resilience* by stripping farmers of the crop diversity and livelihood flexibility that previously buffered them against shocks. Their claim—that CIP households are more exposed to food insecurity when adversity strikes—has not been tested quantitatively. This study addresses that gap.

Theoretical approach. Rwanda's *imihigo* enforcement machinery enabled the state to impose a macro-level productivity goal by removing the micro-level adaptive strategies smallholders use to manage risk. The result is structural vulnerability: CIP households are locked into state-assigned monocrops with reduced ability to adjust cultivation or diversify diets in response to shocks—a cost that falls hardest on those with the fewest alternative resources (savings, credit, off-farm income).

Hypotheses.

H1 Households in districts with higher LUC intensity will show lower dietary diversity scores

when they experience an adverse shock than households in low-intensity districts (*resilience-reducing effect of CIP*).

H2 The vulnerability-amplifying effect of LUC intensity will increase as household wealth decreases (*poverty-concentration hypothesis*).

Operationalization. The key independent variable is **district-level LUC intensity**, constructed as the population-weighted average share of agricultural land under official land consolidation sites across the three seasons of the NISR Seasonal Agricultural Survey (SAS) 2024. For each season, plots are filtered to agricultural land (`s2q7 = 96`), and the LUC share is estimated as the sum of plot area $\times$ plot weight where `s2q12 = 1` divided by total weighted agricultural area, then averaged across seasons. This yields a single annual intensity measure per district. The primary dependent variable is the **household dietary diversity score (DDS)**, constructed from the food consumption module of the EICV7 (2023/24) by collapsing food items into standard groups and counting groups consumed. The **shock variable** is a binary indicator from `s5eq1-s5eq2c` in the EICV7 household module, equal to one if the household reported at least one severe adverse shock in the past 12 months. Household controls (wealth quintile, household size, urban/rural residence) are drawn from the EICV7 poverty file; savings and credit access from `CS_S10C_Savings` and `CS_S10A1_A2_credits` serve as additional controls and potential mediators. The unit of analysis is the **household**, nested within districts ($N \approx 14,000$ households across 30 districts).

Preliminary structure:

1. *From Policy Success to Distributional Question* — Introduction and relevance
2. *What the Literature Misses* — State of the art and the untested resilience claim
3. *Coercion, Specialisation, and Structural Vulnerability* — Theory and hypotheses
4. *Measuring the Cost* — Research design and operationalization
 - 4.1 Data and variables
 - 4.2 Methodology
 - 4.3 Results
5. *Who Bears the Burden?* — Conclusion and implications

Methodology note:

The study employs OLS regression with district fixed effects on the EICV7 cross-sectional household sample, matched to district-level LUC intensity from SAS 2024. The baseline model is:

$$\text{DDS}_{ihd} = \beta_1 \text{LUC}_d + \beta_2 \text{Shock}_{ih} + \beta_3 (\text{LUC}_d \times \text{Shock}_{ih}) + \mathbf{Z}'_{ih} \boldsymbol{\gamma} + \alpha_d + \varepsilon_{ihd} \quad (1)$$

where β_3 is the key coefficient (a negative value supports H1); $\mathbf{Z}_{ih}$ includes household size, wealth quintile, urban/rural residence, and savings/credit access; α_d are district fixed effects; and errors are clustered at the district level. H2 is tested via a triple interaction $\text{LUC}_d \times \text{Shock}_{ih} \times Q_{ih}$, where Q_{ih} is the household consumption quintile.

Data sources:

Variable	Source	Access
Dietary diversity score	EICV7 (2023/24), CS_S8B_Food_Expenditure_Consumption.dta	World Bank Micro-data Catalog; free registration
Shock indicator	EICV7 (2023/24), CS_S01_S5_S7_Household.dta, variables s5eq1–s5eq2c	World Bank Micro-data Catalog; free registration
Household controls & wealth	EICV7, CS_EICV7_poverty_file.dta	Included in EICV7 package
Savings & credit access	EICV7, CS_S10C_Savings.dta; CS_S10A1_A2_credits.dta	Included in EICV7 package
LUC intensity (weighted; s2q12)	NISR Seasonal Agricultural Survey 2024, Seasons A, B & C	NISR microdata catalog; free registration

Literature:

- Ansoms, An, Esther Marijnen, Giuseppe Cioffo, and Jude Murison. 2017. “Statistics versus Livelihoods: Questioning Rwanda’s Pathway out of Poverty.” *Review of African Political Economy* 44(151): 47–65.
- Cantore, Nicola. n.d. *The Crop Intensification Program in Rwanda: A Sustainability Analysis*. London: Overseas Development Institute.
- Cioffo, Giuseppe, An Ansoms, and Jude Murison. 2016. “Modernising Agriculture through a ‘New’ Green Revolution: The Limits of the Crop Intensification Programme in Rwanda.” *Review of African Political Economy* 43(148): 188–204.
- Clay, Nathan, and Karl S. Zimmerer. 2020. “Who Is Resilient in Africa’s Green Revolution? Sustainable Intensification and Climate Smart Agriculture in Rwanda.” *Land Use Policy* 97: 104558.
- Del Prete, Davide, Luc Ghins, Emiliano Magrini, and Karl Pauw. 2019. “Land Consolidation, Specialization and Household Diets: Evidence from Rwanda.” *Food Policy* 83: 139–149.
- Heinen, Sebastian. 2022. “Rwanda’s Agricultural Transformation Revisited: Stagnating Food Production, Systematic Overestimation, and a Flawed Performance Contract System.” *Journal of Development Studies* 58(10): 2044–2064.
- Ingelaere, Bert. 2014. “What’s on a Peasant’s Mind? Experiencing RPF State Reach and Overreach in Post-Genocide Rwanda (2000–10).” *Journal of Eastern African Studies* 8(2): 214–230.
- Matin, Nilufar, John Forrester, and Jonathan Ensor. 2018. “What Is Equitable Resilience?” *World Development* 109: 197–205.
- Nilsson, Pär. 2018. “The Role of Land Use Consolidation in Improving Crop Yields among Farm Households in Rwanda.” *Journal of Development Studies* 55(1): 1–17.
- Nsabimana, Alain, Philip K. Adom, Augustin Mukamugema, and Jean Claude Ngabitsinze. 2023. “The Short and Long Run Effects of Land Use Consolidation Programme on Farm Input

Uptakes: Evidence from Rwanda.” *Land Use Policy* 132: 106787.